INTRODUCTION

In daily life, we use several measuring instruments such as manual balance, litre cans and scale to measure quantities like weight (rice, wheat, etc.), volume of liquid (oil, petrol, diesel, etc.), length (cloth, wood, etc.), respectively. Presently, digital instruments are used, owing to their accurate measurement of quantities with minimum error. Likewise to measure and test electronic components and parameters, we are using several analog and digital instruments.

10.1. TESTING AND MEASURING INSTRUMENTS

Testing and measuring instruments are used to check the electronic components, whether they are in good condition or not and to measure electrical parameters such as current, voltage, resistance, frequency and phase etc. Though, there are numerous measuring instruments in practice, we will see some of the basic and important measuring instruments and their functions in this Chapter.

Some of the basic instruments to be discussed are,

- Multimeter (Analog & Digital)
- Cathode Ray Oscilloscope (C.R.O.)
- Signal Generator
- Function Generator
- Spectrum analyser
- Logic probe
- IC tester
- Digital Energy meter

10.2. MULTIMETER

A multimeter is an electronic instrument which can measure multiple parameters such as current, voltage and resistance.



(i) 0–300 V A.C Voltmeter



(ii) 0–10 V A.C Voltmeter

Figure 10.1 Separate range A.C voltmeters

ELEC_EQUIP_Chapter10.indd 202 11/10/2020 11:55:35 AM